CONFIDENTIAL

SCHOOL OF COMPUTING
Faculty of Engineering

UNIVERSITI TEKNOLOGI MALAYSIA

TEST 2

PROBLEM-SOLVING

SEMESTER II 2020/2021

SUBJECT CODE	**: SECJ2154**
SUBJECT NAME	**: OBJECT ORIENTED PROGRAMMING**
YEAR/COURSE	**: 2 (SECB/ SECJ/ SECP/ SECR/ SECV)**
TIME	**: 10:00 – 12:30 MYT (UTC +8, 150 Minutes)**
DATE/ DAY	**: 5th JUNE 2021 (SATURDAY)**

INSTRUCTIONS TO THE STUDENTS:

- Read the problem and instructions carefully.
- You are given **TWO HOURS THIRTY MINUTES** to complete the test, inclusive of your program's submission (interim and final submissions).

IMPORTANT NOTES:

- All the **COMMENT STATEMENTS** in the submitted program **WILL NOT BE EVALUATED**.

SUBMISSION PROCEDURE:

- Only the source codes' file ***(*.java)*** is required for the submission.
- You do not need to compress the file.
- Submit the source code file via the **UTM's e-learning system**.

This question booklet consists of 7 pages inclusive of the cover page

PROBLEM-SOLVING (100 Marks)

Write **six (6)** complete Java programs named, **`Traveler.java`**, **`TravelerApp.java`** **`Name.java`**, **`PhoneNum.java`**, **`Packages.java`**, and **`Membership.java`** based on the UML class diagram given in **Figure 1**. Your program should be able to produce the output shown in **Figure 2**.

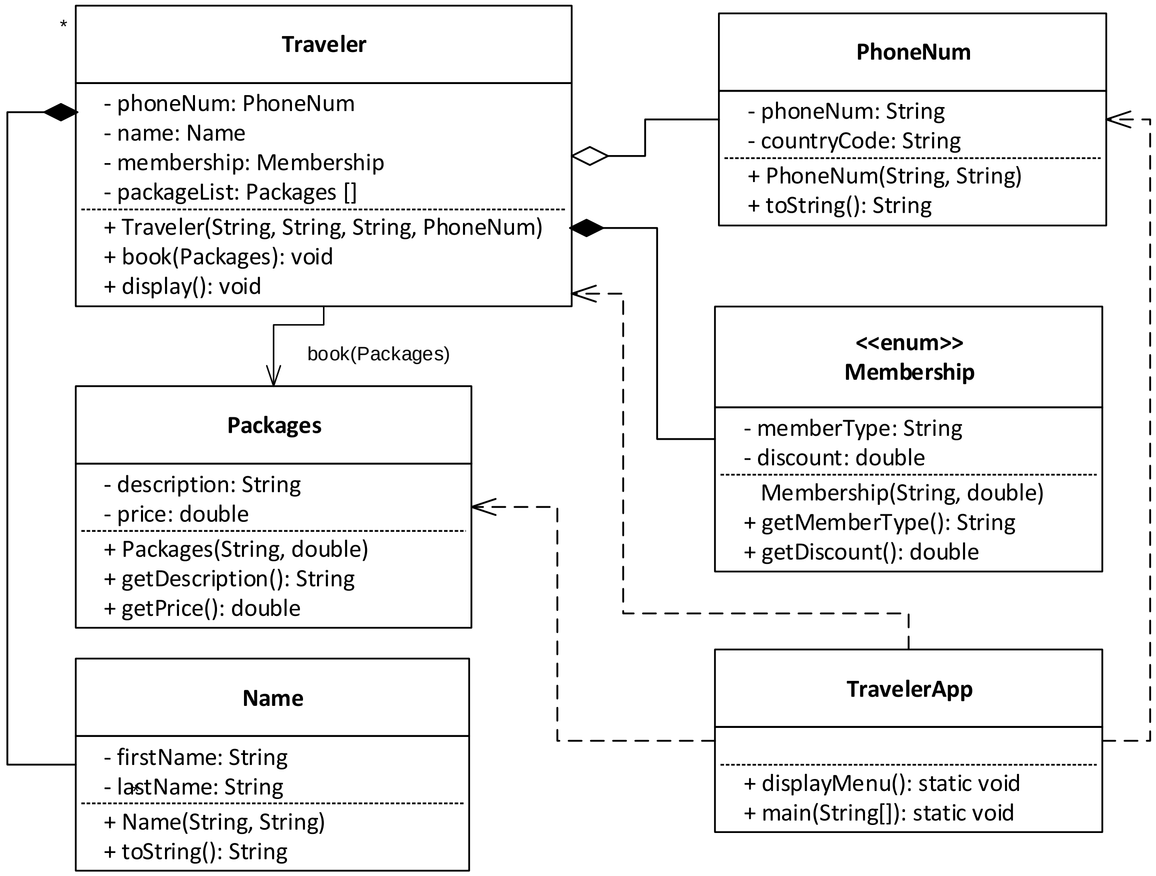

Figure 1: UML class diagram

Implement all the classes with the instance variables (attributes) and methods specified in the diagram. The purpose of each method is as the name implies, and some of them are further explained below. Write the program based on the following tasks:

Task 1: (5.5 Marks)

In **`Packages`** class, do the following tasks:

a) Define all the attributes of the class. (1 mark)

b) Define a constructor with arguments. Initialize all the instance variables for the class with the passed arguments. (2 marks)

c) Define the accessor (getter) methods. (2 marks)

Task 2: (5 Marks)

In **`PhoneNum`** class, do the following tasks:

a) Define all the attributes of the class. (1 mark)

b) Define a constructor with arguments. Initialize all the instance variables for the class with the passed arguments. (2 marks)

c) Define **`toString`** method. The method returns the contact number of the traveler. (1.5 marks)

Task 3: (5 Marks)

In **`Name`** class, do the following tasks:

a) Define all the attributes of the class. (1 mark)

b) Define a constructor with arguments. Initialize all the instance variables for the class with the passed arguments. (2 marks)

c) Define **`toString`** method. The method returns the full name of the traveler. (1.5 marks)

Task 4: (7 Marks)

In **`Membership`** class, do the following tasks:

a) Write the class uses **`enum`** data type. The **`enum`** class has a fixed set of constants. Define the **`enum`** data type based on **all constants** listed in Table 1. (1.5 marks)

Table 1: Set of constants and values for **`Membership`** **`enum`** class

Constants	Seat Class	Discount (%)
S	Silver	5
G	Gold	10
P	Platinum	20

b) Define all the attributes of the class. (1 mark)

c) Define constructor with arguments. Initialize all the attributes for the class with the passed arguments. (2 marks)

d) Define the accessor (getter) methods. (2 marks)

Task 5: (28.5 Marks)

In **`Traveler`** class, do the following tasks:

a) Define all the attributes of the class. **Note:** You need to use **`Vector`** (dynamic array) to declare **`packageList`** attribute. (2.5 marks)

b) Define constructor with arguments. Initialize **`phoneNum`** attribute with a passed **`PhoneNum`** argument. Create the instance of **`Name`** and initialize all the attributes of the instance with passed arguments. Create **`Membership enum`** from a passed **`String`** argument. Finally, create the object from the class **`Vector`** that is defined in Task 5(a). (7 marks)

c) Define **`book`** method. The method is used to add the object from **`Packages`** class to **`packageList`** array. The object added to the array refers to the package booked by a traveler. (1.5 marks)

d) Define **`display`** method. The method displays the traveler's name, contact number, and type of membership, the number of packages, the list of packages, the total price, the discount received, and the price after the discount. The discount given for all packages is depending on the membership type of the traveler. The list of packages must contain the following information: package's description and price. **Figure 2** shows the sample of the output that your program should produce. **Note:** You must use proper output formatting to generate the output. Note also that the **bold** texts indicate input entered by the user. (17 marks)

```
------------------------------------------
   Traveling Package Ordering System
------------------------------------------
[1] Add Traveler
[2] Display Travelers
[3] Exit
Your choice: 1

##### Add Traveler #####
First name: Aishah
Last name: Ali
Contact Number:
   Country code: 60
   Phone Number: 127896541
Membership [S-Silver | G-Gold | P-Platinum]: G
Number of packages: 3
   Package #1:
        Description: Langkawi Mangrove Kayak Tour
        Price: RM 230
   Package #2:
        Description: Langkawi Island Hopping
        Price: RM 45
   Package #3:
        Description: Pulau Payar Snorkeling Beach Tour
        Price: RM 160

------------------------------------------
   Traveling Package Ordering System
------------------------------------------
[1] Add Traveler
[2] Display Travelers
[3] Exit
```

```
Your choice: 1

##### Add Traveler #####
First name: Hamidah
Last name: Abdul Hamid
Contact Number:
   Country code: 60
   Phone Number: 1178951230
Membership [S-Silver | G-Gold | P-Platinum]: p
Number of packages: 2
   Package #1:
        Description: Danna Hotel Honeymoon Package
        Price: RM 1440
   Package #2:
        Description: Langkawi Sunset Dinner Cruise
        Price: RM 250

------------------------------------------
   Traveling Package Ordering System
------------------------------------------
[1] Add Traveler
[2] Display Travelers
[3] Exit
Your choice: 2

##### List of Travelers #####

Traveler #1
Name: Aishah Ali
Contact No.: +60-127896541
Membership: Gold
Number of packages: 3
1)  Langkawi Mangrove Kayak Tour            RM  230.00
2)  Langkawi Island Hopping                 RM   45.00
3)  Pulau Payar Snorkeling Beach Tour       RM  160.00

Total price: RM 435.00
Discount: RM 43.50
Price after discount: RM 391.50

Traveler #2
Name: Hamidah Abdul Hamid
Contact No.: +60-1178951230
Membership: Platinum
Number of packages: 2
1)  Danna Hotel Honeymoon Package           RM 1440.00
2)  Langkawi Sunset Dinner Cruise           RM  250.00

Total price: RM 1690.00
Discount: RM 338.00
Price after discount: RM 1352.00

------------------------------------------
   Traveling Package Ordering System
------------------------------------------
[1] Add Traveler
[2] Display Travelers
[3] Exit
Your choice: 3

Thank you for using this system :)
```

Figure 2: Example output of a program

Task 6: (43 Marks)

In **`TravelerApp`** class, do the following tasks:

a) Define **`displayMenu`** method to provide the user a menu-driven interaction. The definition for the **`displayMenu`** method is fully given in **Figure 3**.

```
public static void displayMenu() {
  System.out.println("------------------------------------------");
  System.out.println("   Traveling Package Ordering System");
  System.out.println("------------------------------------------");
  System.out.println("[1] Add Traveler");
  System.out.println("[2] Display Travelers");
  System.out.println("[3] Exit");
  System.out.print("Your choice: ");
}
```

Figure 3: `displayMenu` function

b) Define **`main`** method with following tasks:

- Create a **`Vector`** of objects from class **`Traveler`** to store the values for **`Traveler`** attributes. (2 marks)
- Define objects from class **`PhoneNum`**, **`Traveler`**, and **`Packages`** and define any suitable variables for the program. You may also need to import a class if necessary. (4 marks)
- Implement **`displayMenu`** to ask the user to key-in choices. (8 marks)

  i) If the choice is **`1`**:

  - Display messages for choice **`1`** as shown in **Figure 2** and read values from the keyboard. Use the values to initialize the object from class **`PhoneNum`**, **`Traveler`**, and **`Packages`**.
  - Get the number of packages that the traveler wants to book and use a loop to get the description and price for the package, and implement **`book`** method for each package. (19 marks)

  ii) If the choice is **`2`**:

  - Display messages for choice **`2`** and as shown in **Figure 2**. Execute **`display`** method for each traveler in the **`Vector`**. (5.5 marks)

  iii) If the choice is **`3`**:

  - Exit from the system and display an appropriate message. (0.5 marks)

Task 7: (6 Marks)

a) Use an appropriate structure for the program:

- Import an appropriate class. (1 mark)
- Use proper output formatting. (1 mark)
- The code is indented correctly. (1 mark)

b) The program is able to run, work, and display the output as required. (3 marks)